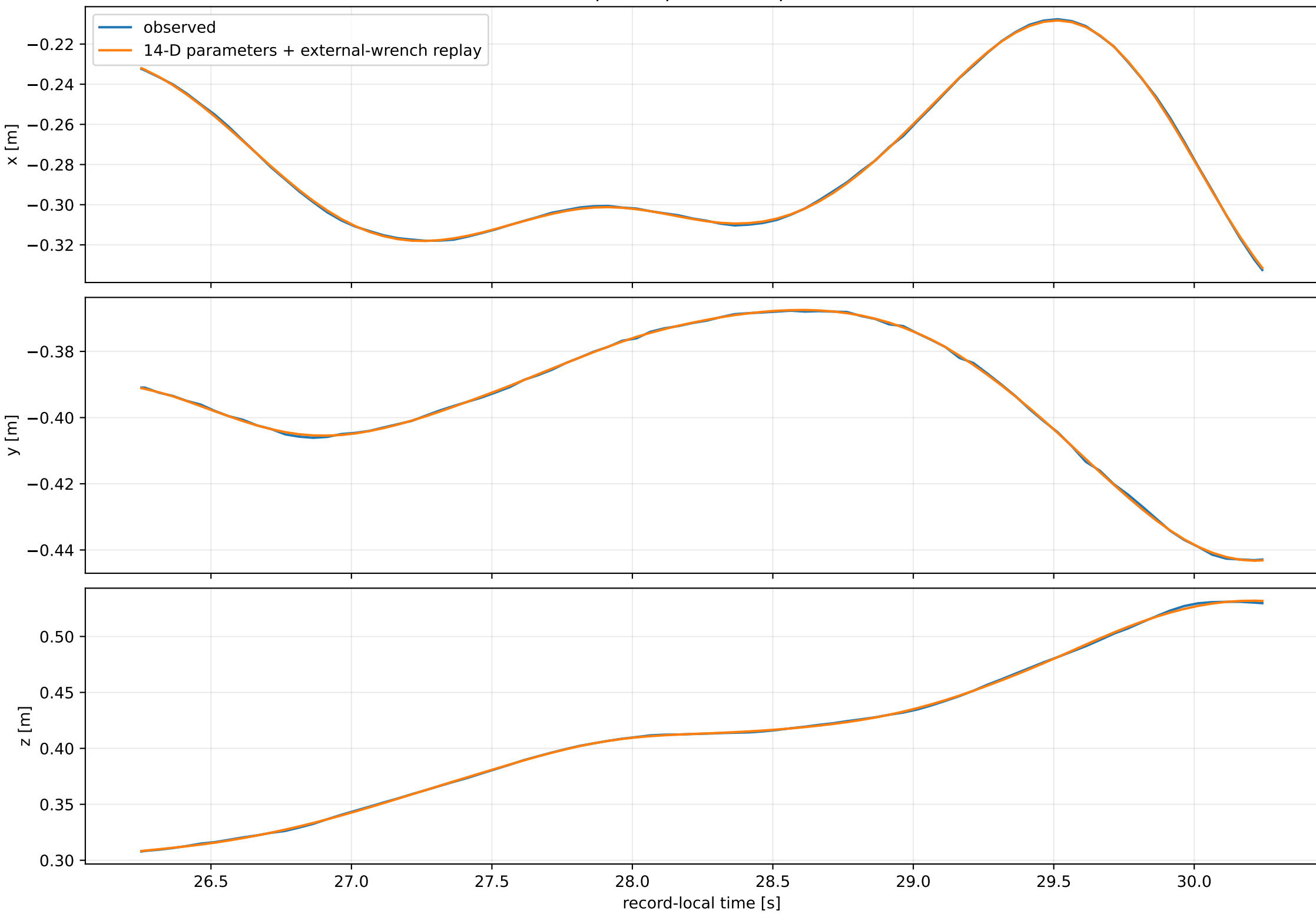
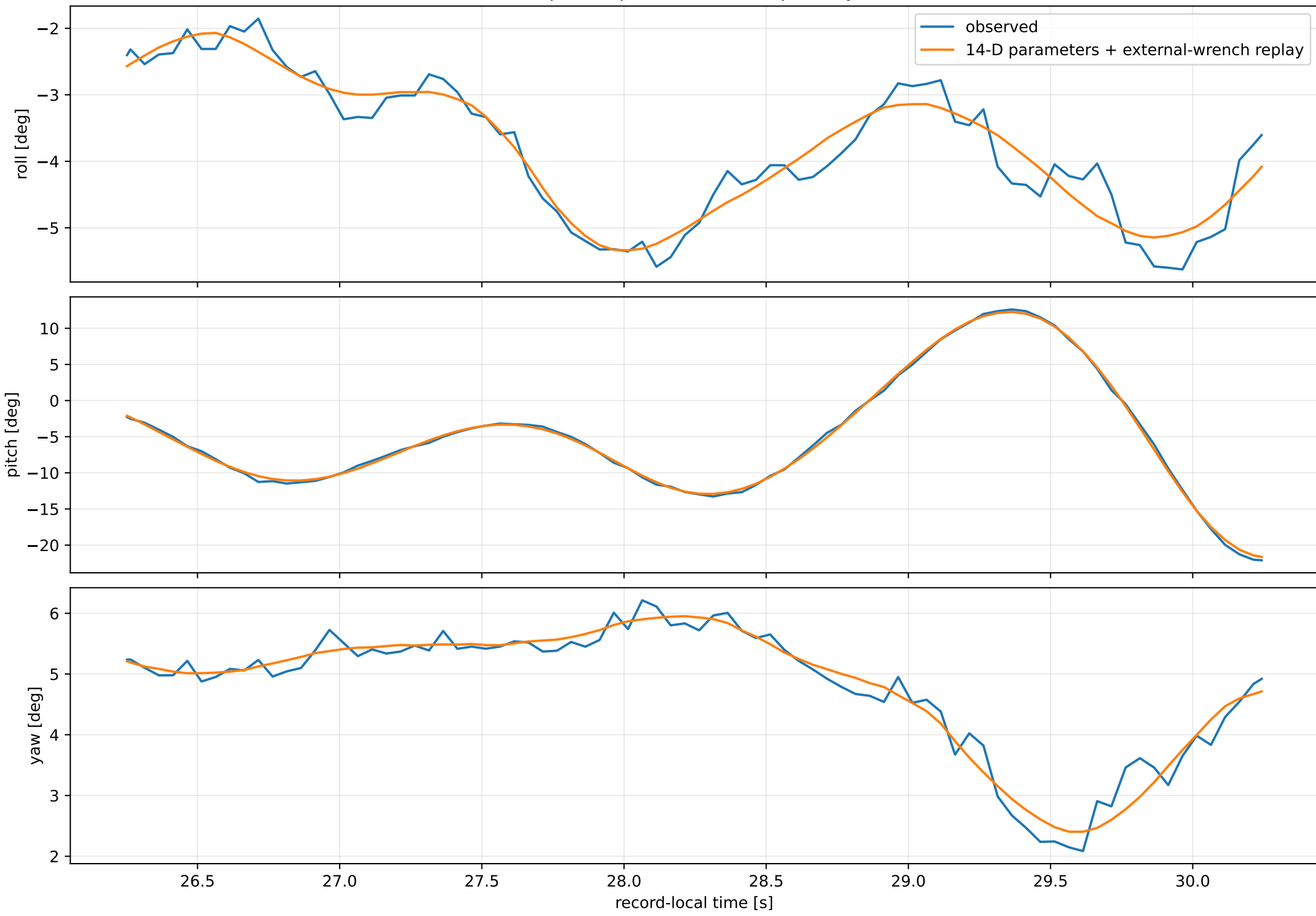


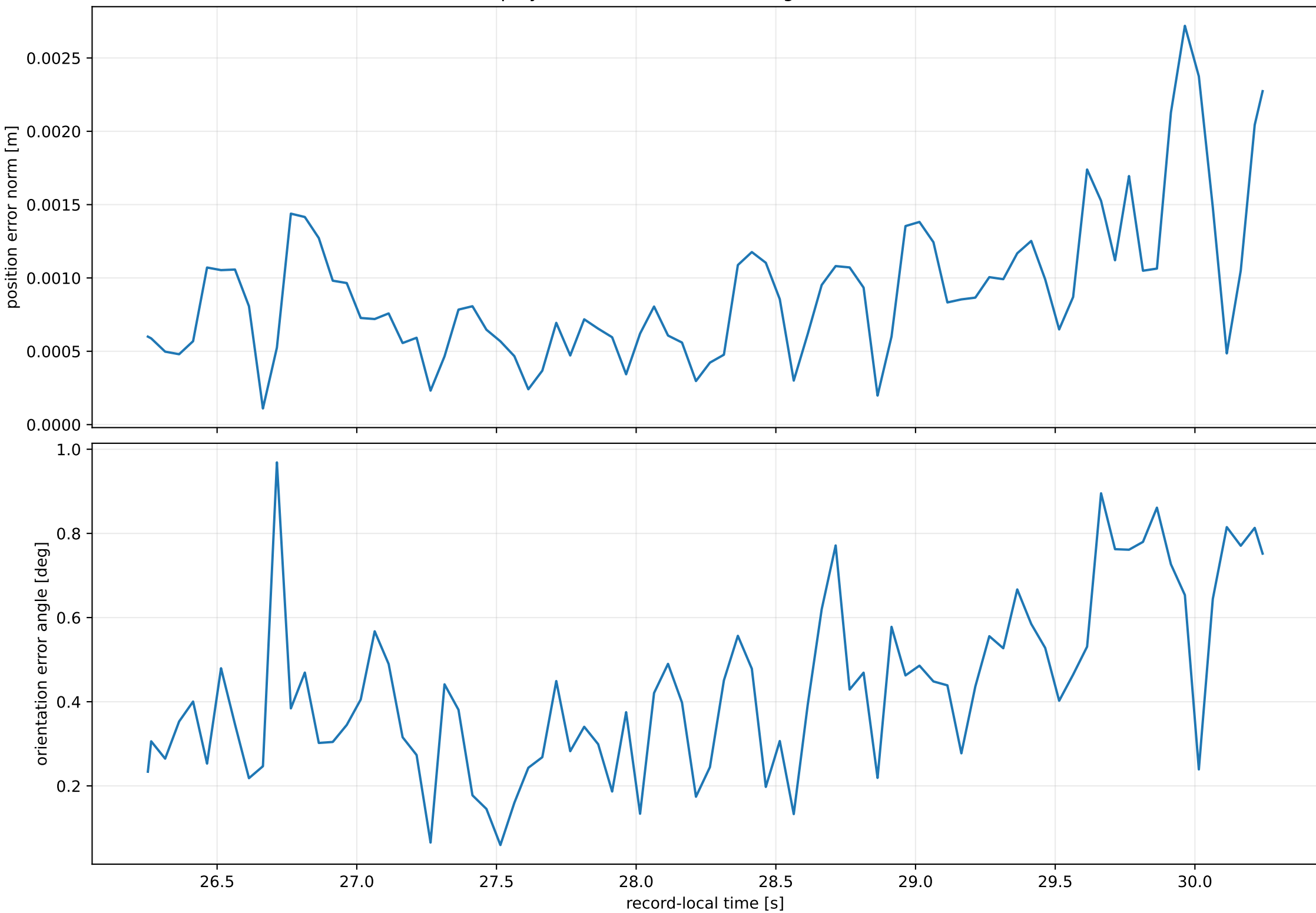
Fine pose reproduction: position



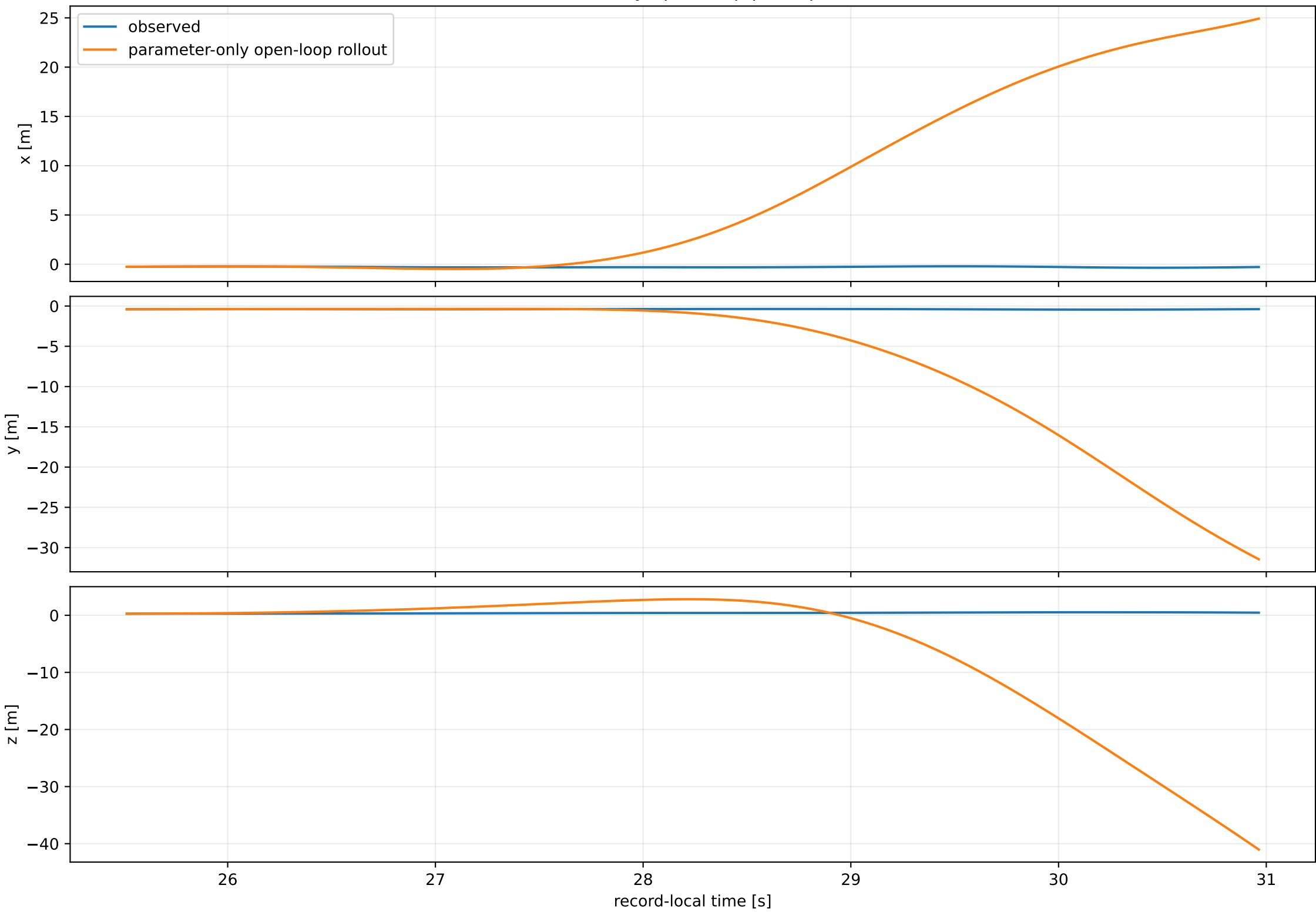
Fine pose reproduction: roll / pitch / yaw



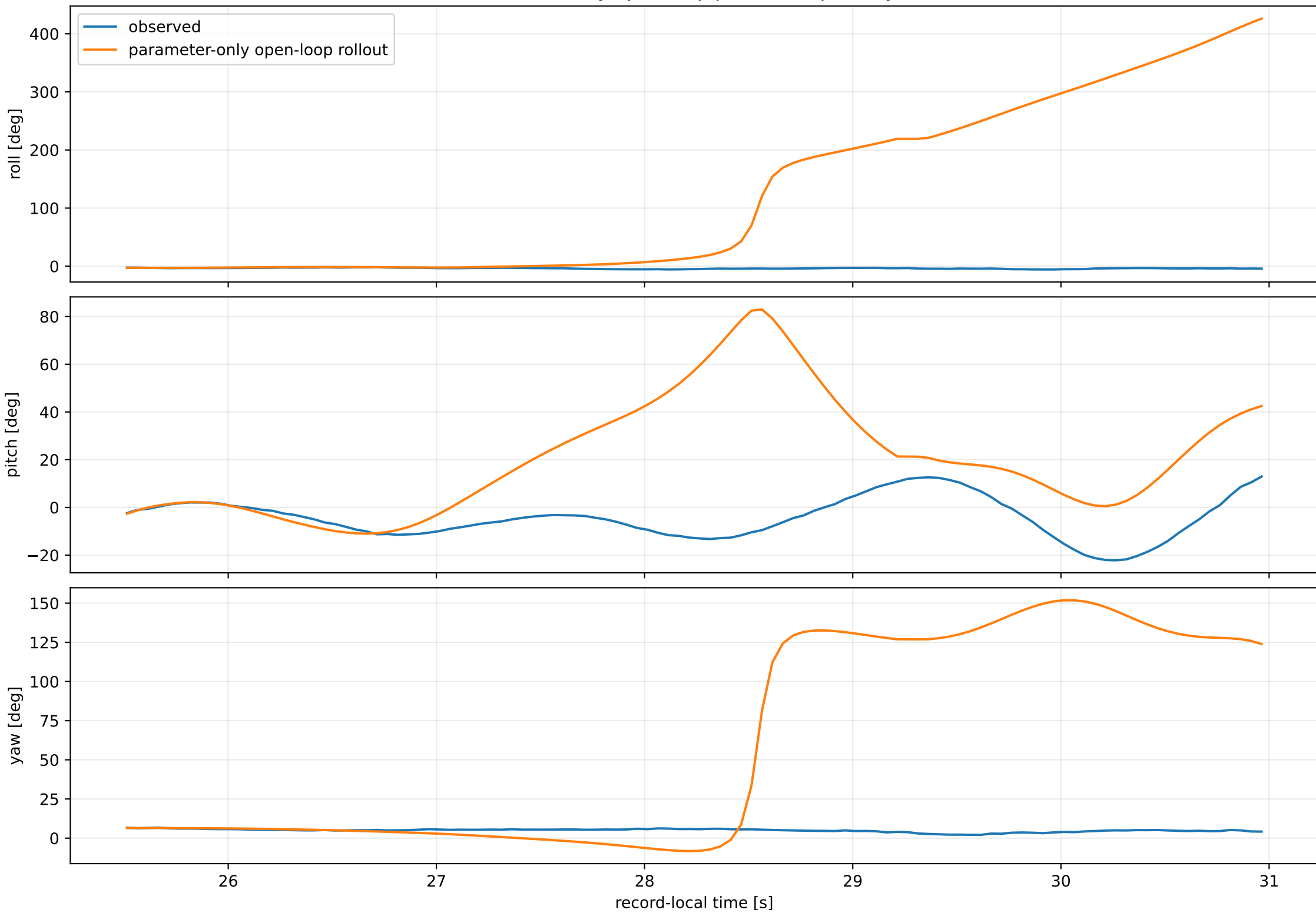
Fine replay error after the 14-D change — lower is better



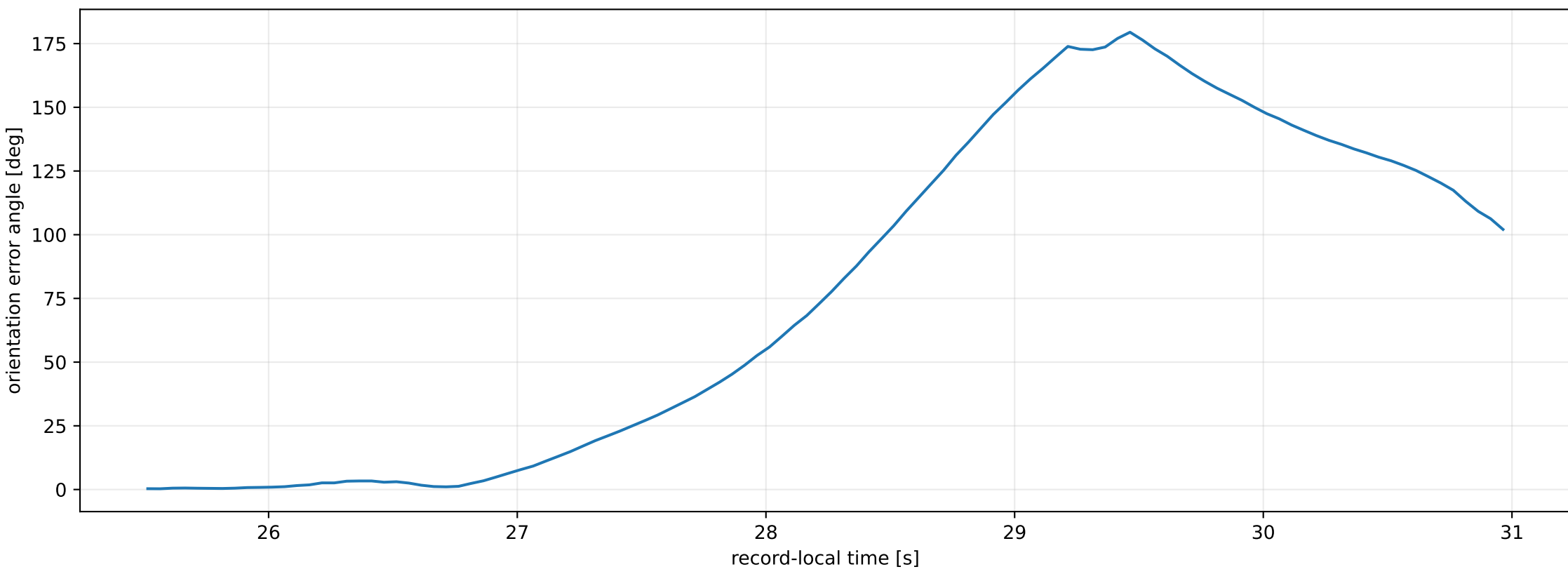
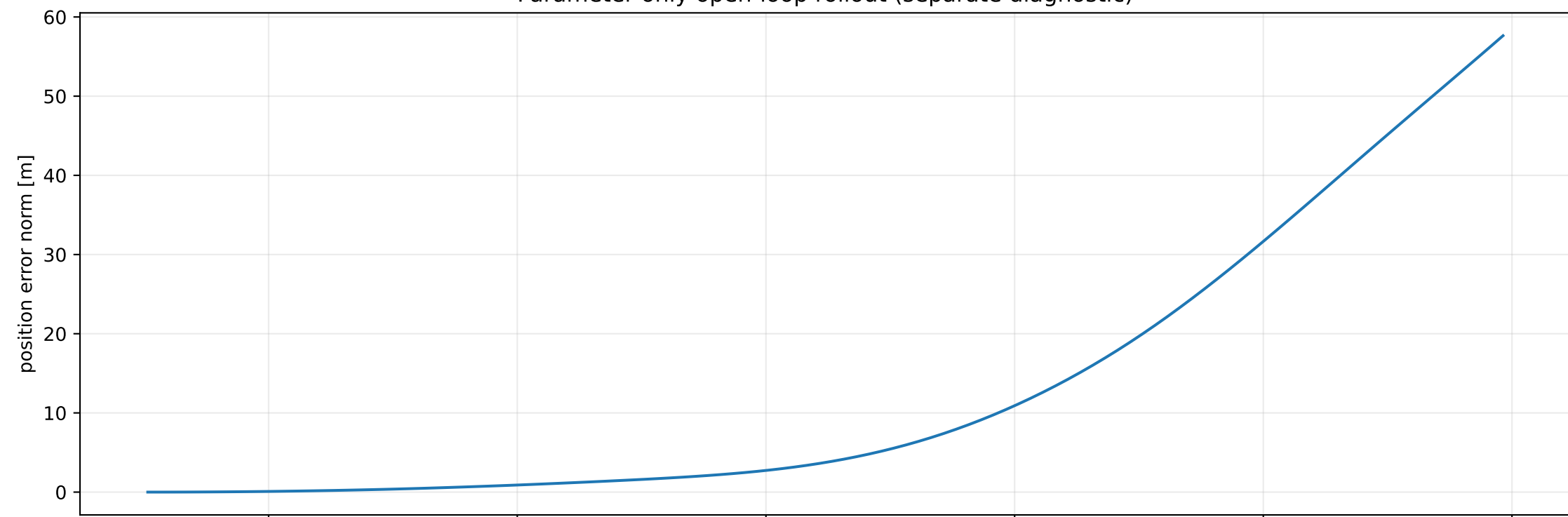
Parameter-only open-loop pose: position



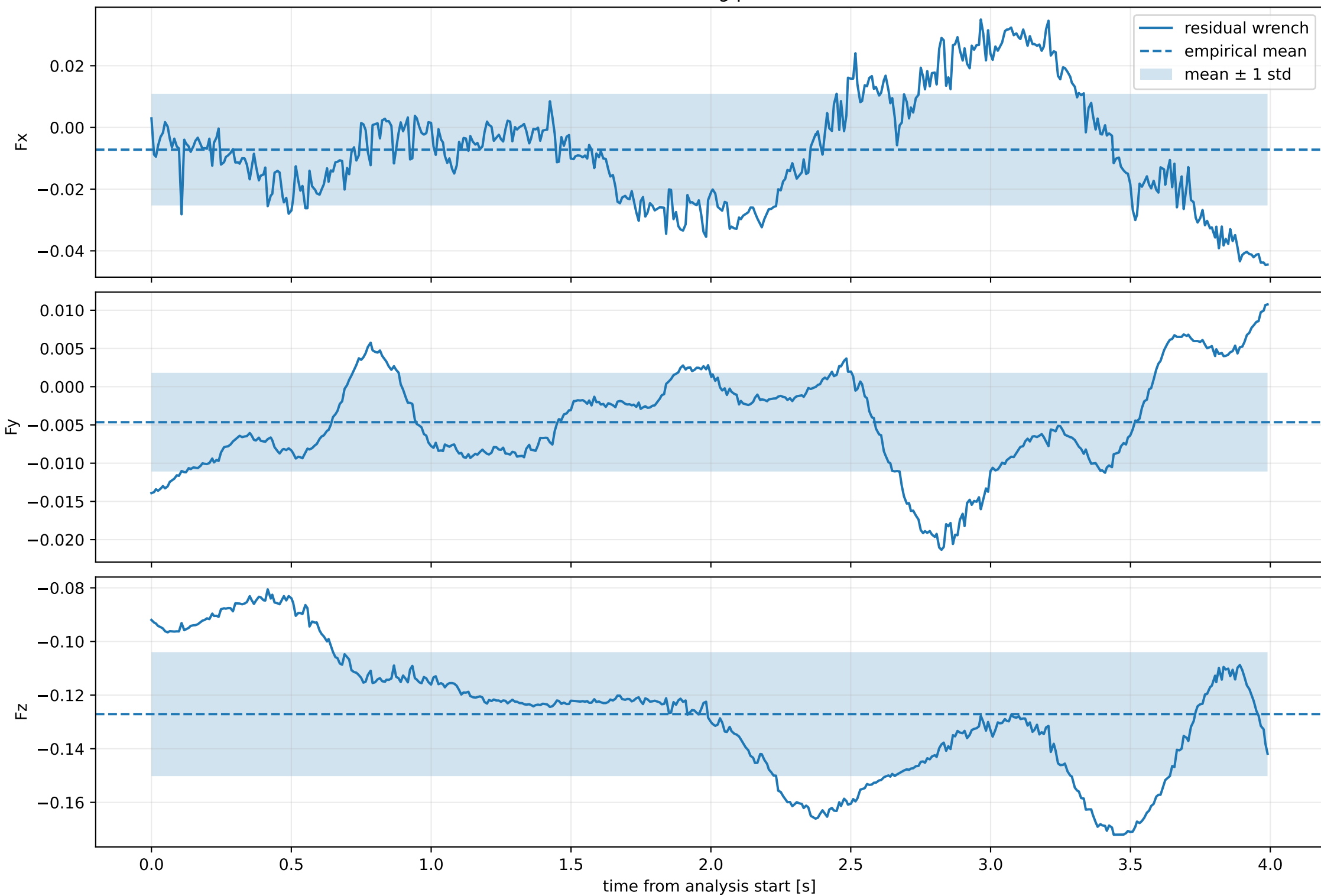
Parameter-only open-loop pose: roll / pitch / yaw



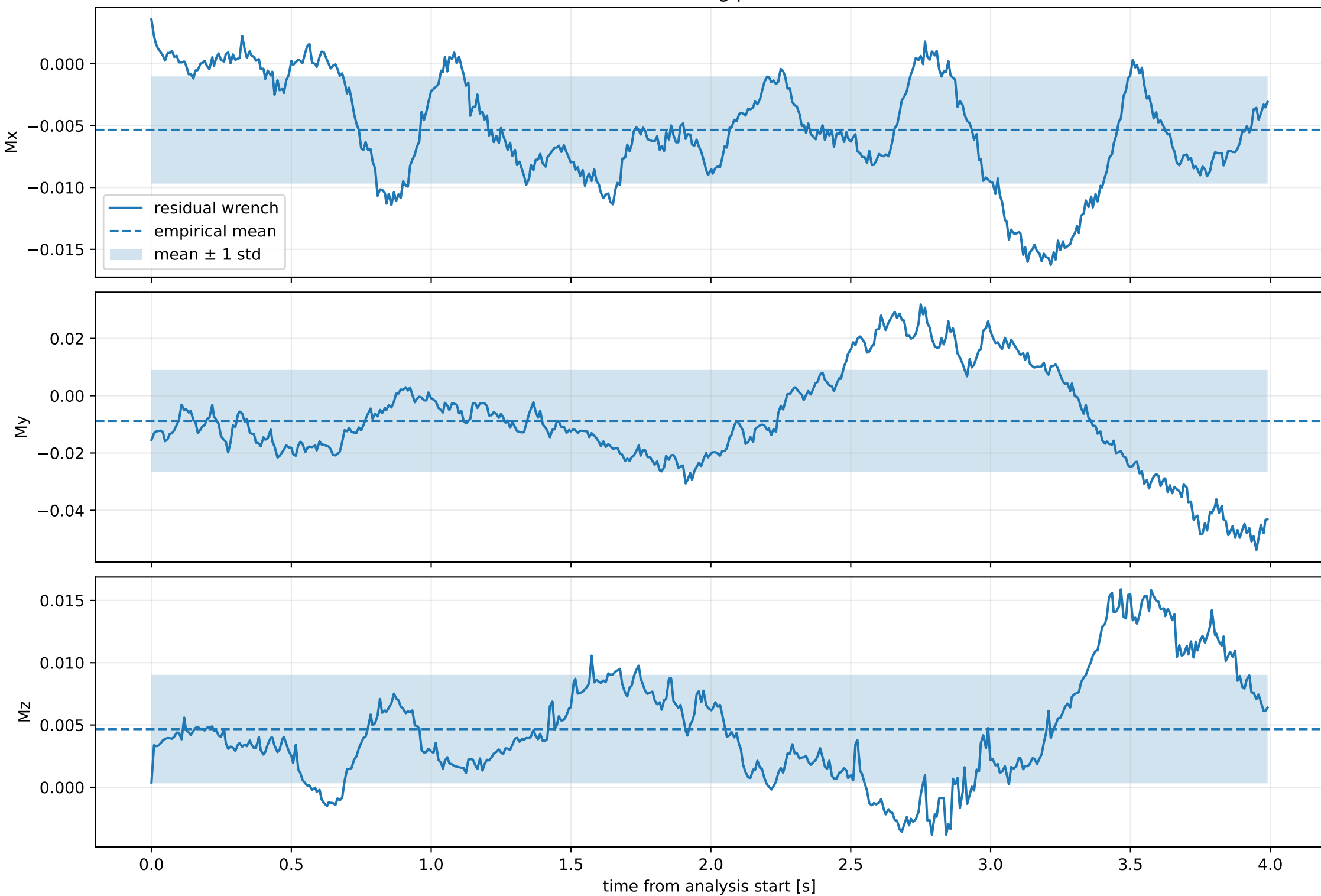
Parameter-only open-loop rollout (separate diagnostic)



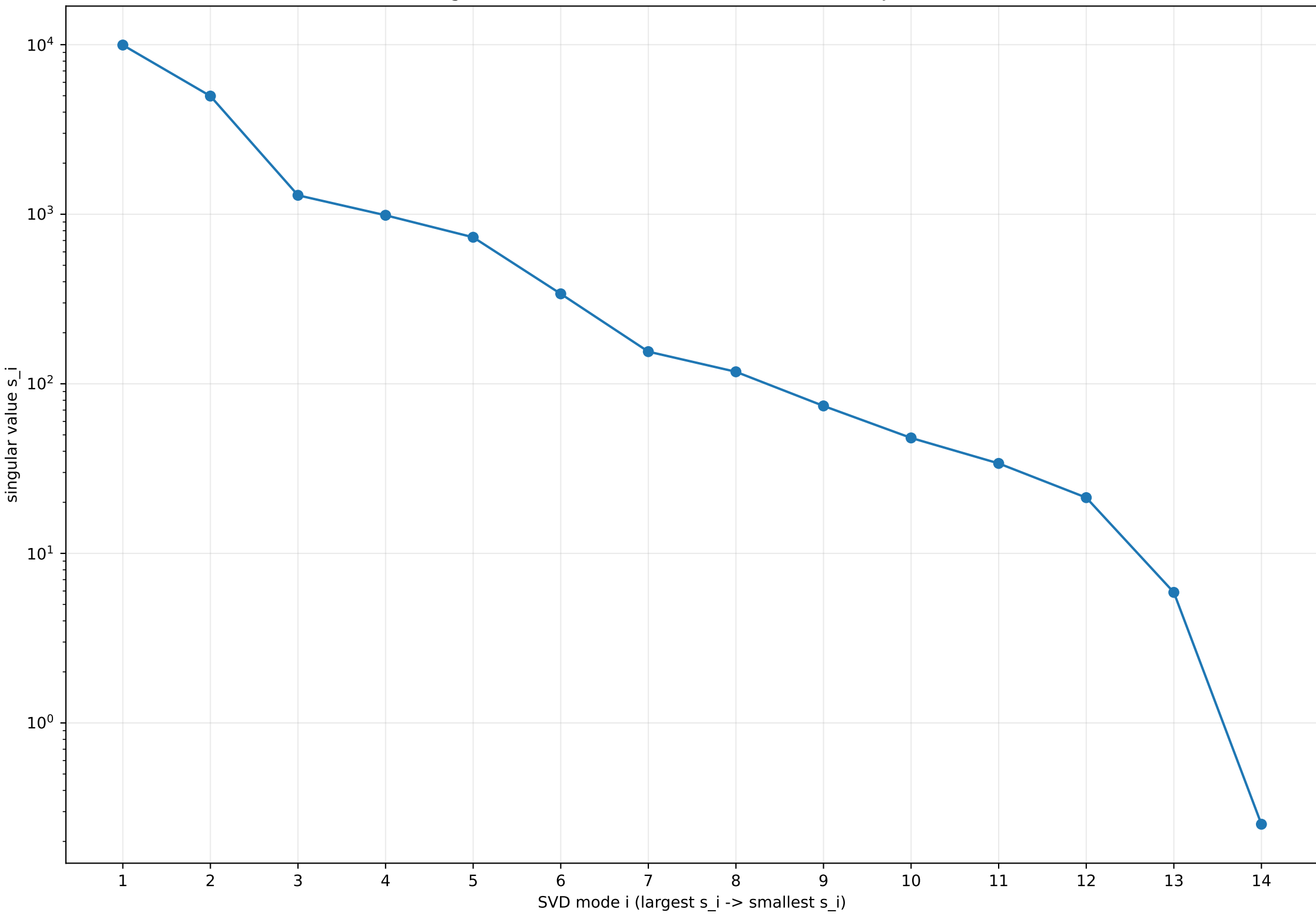
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



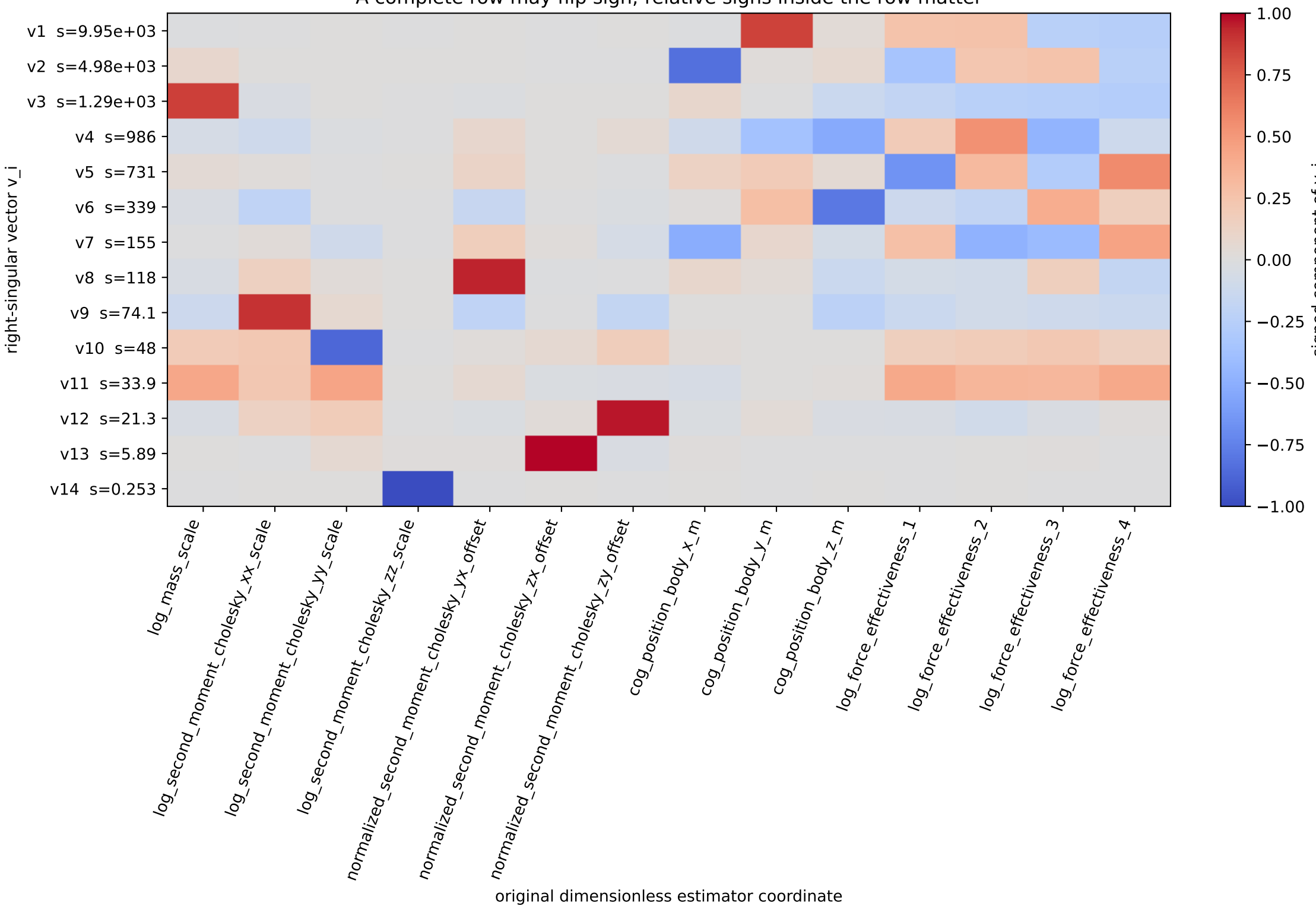
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36618572046

inertia J [kg m²]

[0.065002574 3.5494791e-06 1.7345701e-05]
[3.5494791e-06 0.064984074 -3.1587871e-06]
[1.7345701e-05 -3.1587871e-06 0.12899252]

principal moments [kg m²]

[0.064983415, 0.065003227, 0.12899253]

CoG position in body frame [m]

[-0.0026415224, 0.00038737055, 0.0097157292]

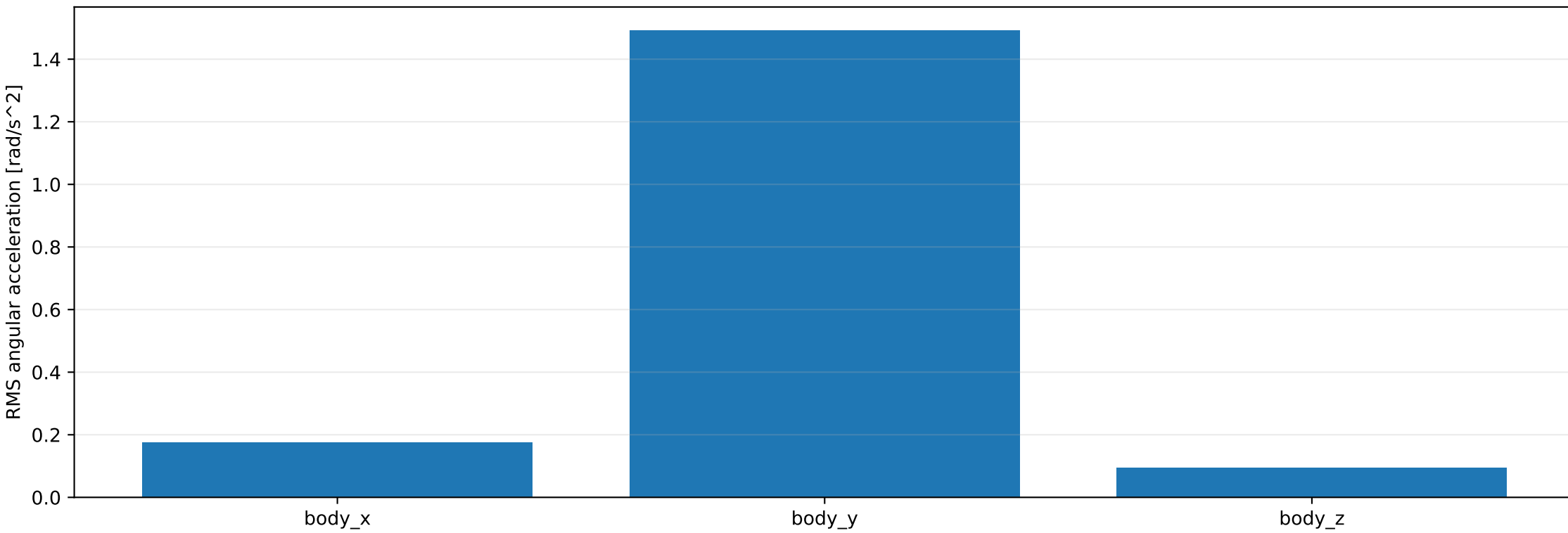
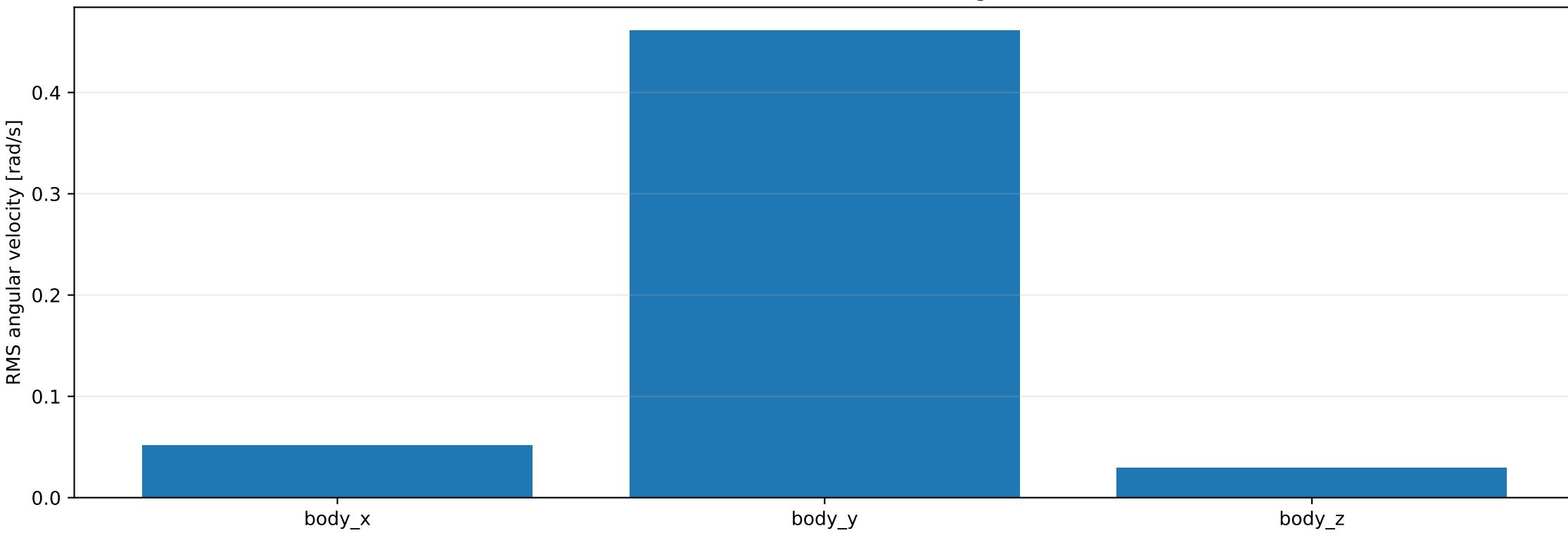
rotor force effectiveness

[0.96221626, 0.97132316, 0.96572554, 0.95643635]

selected delay [s]

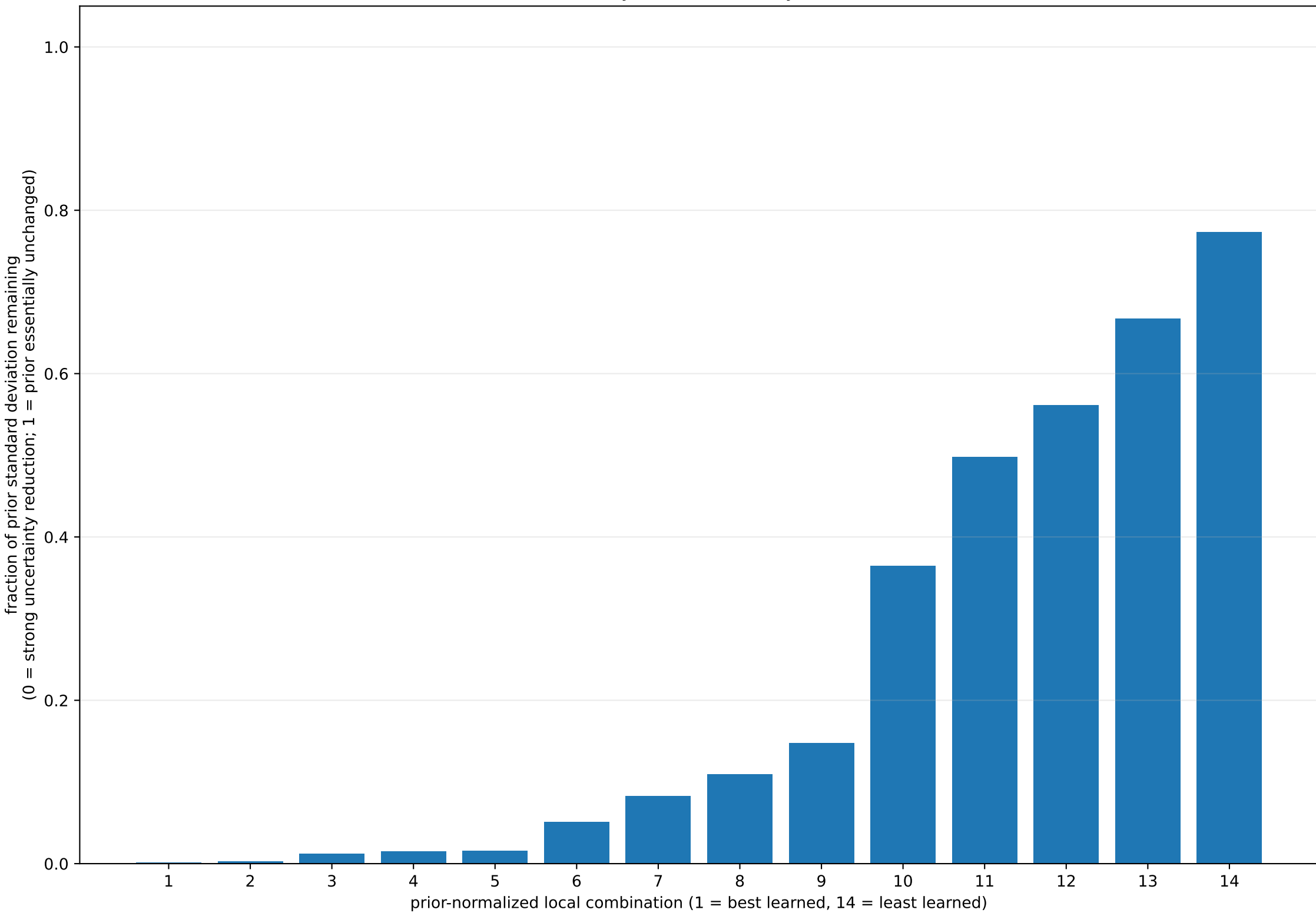
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

